Contents

Soft Mutation in Colloquial Welsh: A Predictive Account

Scope. Prescriptive standard modern **colloquial** Welsh, with Gareth King's *Modern Welsh: A Comprehensive Grammar* (3rd ed., 2016) as the normative reference. The account is strictly **predictive**: given a word and its grammatical context, does it exhibit soft mutation (SM)? Real spoken usage is not categorical (Jones, n.d.; Knight et al. 2020) — §7 records where prescription and usage genuinely diverge. The phonological *content* of mutation (t→d, p→b, c→g, …) is taken as opaque throughout; only the *trigger conditions* are at issue. Word-internal mutation under derivational prefixes is out of scope: prefixed forms are radical lexemes in their own right.

1. The heterogeneity thesis

Every current theoretical account, whatever its framework, starts from the same premise: soft mutation has no single cause. Most instances have a lexical or morphosyntactic trigger, while a distinct residue arises under specifically syntactic conditions (Tallerman 2006). Green (2003) states the consensus most bluntly: trigger environments split into proclitic-triggered and syntax-triggered classes, and *no single generalization covers both* — the full set is arbitrary, partly item-specific, and dialectally variable. Historically this arbitrariness is the residue of reanalysis: mutation was once predictable from pure phonology (the final segment of the preceding word), but by Middle Welsh it could be predicted only from a list of triggering environments, the overwhelming majority being individual lexical items acting on the immediately following word (Willis 2007).

Soft mutation accordingly decomposes into three subsystems, plus two general exemptions; every mutation below is attributed to exactly the rule that licenses it (rule labels in parentheses):

1. **Contact triggers** (`lex:*`) — a closed class of words governing a mutation grade on the immediately following dependent.
2. **Gender marking** (`gend:*`) — feminine singular nouns after the article and *un*; modifiers agreeing with a feminine singular controller.

3. **Syntactic/positional mutation** (`synt:*`) — the XP Trigger Hypothesis plus a small set of designated positions.

One constraint spans all three — the **Trigger Constraint** (Breit 2019): every mutation is licensed by the immediately preceding string element or by a feature borne by the target itself. Nothing acts at a distance. This is the strongest cross-cutting invariant in the literature, and every rule below respects it: each conditions only on the immediately preceding element and on the word's own resolved features.

2. Reading the analyses

Each example shows a constituent analysis; numbers on the branches index a node's daughters. Every word is annotated with its outcome: `→ SM` names the licensing rule; `→ radical` names the reason — `no-license` (no rule is in force at that position), `veto:no-reflex blocks …` (a rule is in force, but the word's initial has no soft-mutation reflex; the blocked rule is named), or `veto:immutable blocks …` (the word is lexically immutable; the blocked rule is named). `⟨N f sg⟩` gives category and features; `lemma=` distinguishes homographs (the two *yn*, the two *ei*); `gap:NP` is an extraction site. In the Welsh text, ° marks a soft-mutated word. Annotations appear on *every* word, so incidental facts (the colloquial mutation of the clause-initial verb, exemptions on function words) are visible alongside each example's point.

3. Contact triggers

The majority case (Willis 2007). Each trigger is a lexical entry governing a grade: full SM; *limited* SM sparing *ll-/rh-* (the article, *un*, predicative *yn*, *mor* — King §9); AM; NM; the *mixed* grade (AM on *c/p/t*, SM elsewhere — King §10); or nothing (absence is data: *pob*, *mewn*, *rhwng*, *efo*, *tair* are recorded as non-triggers rather than omitted). AM and NM triggers matter to an account of soft mutation because they *pre-empt*: a word governed by an AM trigger does not soft-mutate. Blocking (King §5d) requires no separate statement — a trigger acts only on the immediately following word, so an intervener simply *is* the preceding element.

i °dŷ
'to a house' — the preposition i° governs SM on its dependent (King §460)

```
PP
├─0 i ⟨Other⟩ → radical (no-license)
└─1 NP
   └─0 tŷ ⟨N m sg⟩ → SM (lex:i)
```

dy °gath
'your cat' — 2sg possessive dy° (King §111)

```
NP
├─0 dy ⟨Other⟩ immutable → radical (no-license)
└─1 cath ⟨N f sg⟩ → SM (lex:dy)
```

ei °gath e — ei chath hi
'his cat' (SM) vs 'her cat' (AM — outside soft mutation) — one orthographic word, two lemmas, two grades (King §112)

```
NP
├─0 ei ⟨Other⟩ lemma=ei.3sgm → radical (no-license)
├─1 cath ⟨N f sg⟩ → SM (lex:ei.3sgm)
└─2 e ⟨Other⟩ → radical (no-license)
```

> As `ei.3sgf` the governed grade is AM: the same position shows `radical (no-license)` — ‘her’ never soft-mutates.

fy nghath
‘my cat’ — fy governs NM, so soft mutation is correctly withheld (King §110)

```
NP
├─0 fy ⟨Other⟩ → radical (no-license)
└─1 cath ⟨N f sg⟩ → radical (no-license)
```

mor °fach — mor llawn
‘so small’ vs ‘so full’ — mor° is limited SM: it spares ll-/rh- (King §105e)

```
AP
├─0 mor ⟨Adv⟩ immutable → radical (no-license)
└─1 bach ⟨Adj⟩ → SM (lex:mor)
```

> Substituting llawn (ll-) at the same position yields `radical (no-license)`: the SM-ltd grade has no reflex for ll-/rh-.

mor llawn
‘so full’ — the ll- target resists the limited-SM trigger

```
AP
├─0 mor ⟨Adv⟩ immutable → radical (no-license)
└─1 llawn ⟨Adj⟩ → radical (no-license)
```

dwy °gath — tair cath
‘two cats’ vs ‘three cats’ — dwy° governs SM, tair governs nothing (King §162)

```
NP
├─0 dwy ⟨Num⟩ → radical (no-license)
└─1 cath ⟨N f sg⟩ → SM (lex:dwy)
```

tair cath
‘three cats’ — feminine tair is always followed by the radical (King §162c)

```
NP
├─0 tair ⟨Num⟩ → radical (no-license)
└─1 cath ⟨N f sg⟩ → radical (no-license)
```

chwe cheffyl
‘six horses’ — chwe governs AM, never SM (King §162e)

```
NP
├─0 chwe ⟨Num⟩ → radical (no-license)
└─1 ceffyl ⟨N m sg⟩ → radical (no-license)
```

chwe mlynedd — dwy °flynedd
‘six years’ (NM via the year-word frame) vs ‘two years’ (plain SM) — one lemma, two frames (King §176)

```
NP
├─0 chwe ⟨Num⟩ → radical (no-license)
└─1 blynedd ⟨N f sg⟩ → radical (no-license)
```

> chwe governs two distinct conditions: AM generally, NM restricted to blwydd/blynedd/diwrnod. Neither yields SM. dwy °flynedd (example above) shows the same noun soft-mutating after a genuine SM trigger.

hen °ddyn
‘an old man’ — prenominal adjectives govern SM (King §96)

```
NP
├─0 hen ⟨Adj⟩ → radical (no-license)
└─1 dyn ⟨N m sg⟩ → SM (lex:hen)
```

pob dyn
‘every man’ — pob is the one prenominal adjective that governs nothing (King §97)

```
NP
├─0 pob ⟨Other⟩ → radical (no-license)
└─1 dyn ⟨N m sg⟩ → radical (no-license)
```

cath neu °gi
‘a cat or a dog’ — neu° governs SM on the second conjunct (King §512)

```
NP
├─0 NP
│  └─0 cath ⟨N f sg⟩ → radical (no-license)
├─1 neu ⟨Other⟩ → radical (no-license)
└─2 NP
   └─0 ci ⟨N m sg⟩ → SM (lex:neu)
```

y °gath neu’r ci
‘the cat or the dog’ — the article ’r intervenes and blocking follows from string adjacency (King §5d)

```
NP
├─0 NP
│  ├─0 y ⟨Other⟩ → radical (no-license)
│  └─1 cath ⟨N f sg⟩ → SM (gend:art-fem-sg)
├─1 neu ⟨Other⟩ → radical (no-license)
└─2 NP
   ├─0 y ⟨Other⟩ → radical (veto:no-reflex blocks lex:neu)
   └─1 ci ⟨N m sg⟩ → radical (no-license)
```

> No blocking statement is needed: ’r simply IS the preceding element, and the condition it governs (feminine singular nouns) does not extend to masculine ci.

4. The gender subsystem

Welsh marks gender by mutation rather than affixation. Two environments: the definite article (and *un*) mutates feminine singular nouns, sparing *ll-/rh-* (King §28); and modifiers of a feminine singular controller mutate — with *no ll-/rh-* sparing (King §100: *profedigaeth °lem*), so the carve-out is category-sensitive, not phonological (Green 2003). Agreement is analysed as a feature borne by the target (Breit 2019), not as adjacency to the noun: that is what carries adjective chains, and what lets a *prenominal* adjective agree with a head that follows it. Possessor phrases are systematically immune despite string adjacency (Mittendorf & Sadler 2006; Dowle 2024) — the immunity is the *absence of any licensing relation*, a consequence of the genitive configuration, not an exemption.

y °gath — y llong — y plant
'the cat / the ship / the children' — the article mutates feminine singulars only, sparing ll-/rh- (King §28)

```
NP
├─0 y ⟨Other⟩ → radical (no-license)
└─1 cath ⟨N f sg⟩ → SM (gend:art-fem-sg)
```

y llong
'the ship' — feminine singular in ll- resists the article (King §28 note b)

```
NP
├─0 y ⟨Other⟩ → radical (no-license)
└─1 llong ⟨N f sg⟩ → radical (no-license)
```

y cathod
'the cats' — feminine plurals pattern with masculines: no mutation (King §28)

```
NP
├─0 y ⟨Other⟩ → radical (no-license)
└─1 cathod ⟨N f pl⟩ → radical (no-license)
```

y °ddau °gi
'the two dogs, both dogs' — the article's second frame mutates the numerals dau/dwy, which then trigger in turn (King §29)

```
NP
├─0 y ⟨Other⟩ → radical (no-license)
├─1 dau ⟨Num⟩ → SM (lex:y)
└─2 ci ⟨N m sg⟩ → SM (lex:dau)
```

cath °fach
'a little cat' — adjectives agree with a feminine singular controller (King §102)

```
NP
├─0 cath ⟨N f sg⟩ → radical (no-license)
└─1 AP
   └─0 bach ⟨Adj⟩ → SM (gend:agr-mod)
```

y °ferch °fach °wen
'the little white girl' — the second adjective is not adjacent to the noun, yet mutates: agreement is a feature borne by the target (Breit 2019), not a contact effect

```
NP
├─0 y ⟨Other⟩ → radical (no-license)
├─1 merch ⟨N f sg⟩ → SM (gend:art-fem-sg)
├─2 AP
│  └─0 bach ⟨Adj⟩ → SM (gend:agr-mod)
```

```
└─3 AP
   └─0 gwyn ⟨Adj⟩ → SM (gend:agr-mod)
```

ci mawr coch
'a big red dog' — masculine chains show no mutation anywhere; this datum forces the NP-internal exclusion on the syntactic subsystem (see §5)

```
NP
├─0 ci ⟨N m sg⟩ → radical (no-license)
├─1 AP
│  └─0 mawr ⟨Adj⟩ → radical (no-license)
└─2 AP
   └─0 coch ⟨Adj⟩ → radical (no-license)
```

y °brif °ddinas
'the capital city' — the PREnominal adjective agrees with a FOLLOWING feminine head; agreement looks rightward within the NP

```
NP
├─0 y ⟨Other⟩ → radical (no-license)
├─1 prif ⟨Adj⟩ → SM (gend:agr-mod)
└─2 dinas ⟨N f sg⟩ → SM (lex:prif)
```

cath merch
'a girl's cat' — the possessor is immune despite string adjacency to a feminine noun (Mittendorf & Sadler 2006; Dowle 2024)

```
NP
├─0 cath ⟨N f sg⟩ → radical (no-license)
└─1 NP
   └─0 merch ⟨N f sg⟩ → radical (no-license)
```

> Derived, not stipulated: in the genitive configuration [NP N NP] the second nominal stands in the possessor relation, and possessors fall outside every licensing relation.

canol y °dre
'the town centre' — immunity applies to the possessor boundary, not to the possessor's inside: the article still fires within it

```
NP
├─0 canol ⟨N m sg⟩ → radical (no-license)
└─1 NP
   ├─0 y ⟨Other⟩ → radical (no-license)
   └─1 tre ⟨N f sg⟩ → SM (gend:art-fem-sg)
```

5. Syntactic and positional mutation

The theoretically contested residue. Two families competed: Case-based accounts, on which direct object mutation (DOM) realizes accusative Case (Zwicky 1984; Roberts 1997, 2005), and the configurational **XP Trigger Hypothesis** (Harlow 1989; Borsley 1997; Tallerman 2006), on which any phrasal category triggers SM on the word immediately following its right edge. The Case account is generally judged to both overgenerate and undergenerate (Borsley 1997; Tallerman 2006): mutation after subjects hits non-accusative material (*dim*, adverbs), and objects of nonfinite verbs mutate exactly when a phrase intervenes (Green 2003) — facts the XPTH derives for free. Stated geometrically: a phrasal constituent (never a clause) whose right edge immediately precedes the target and c-commands it licenses `synt:xp-edge`. Extraction gaps count as phrases (Willis 2007 documents the transparency lineage); following Dowle (2024), no other empty categories are posited.

Two refinements are forced by the data. First, the raw geometry overgenerates **NP-internally**: in a masculine chain like *ci mawr coch*, AP(*mawr*) ends immediately before *coch* and c-commands it, yet *coch* is radical. The rule therefore does not operate between sisters within a noun phrase — Tallerman's 'complement' condition stated configurationally, and exactly Breit's (2019) division of labor: NP-internal mutation belongs to the gender subsystem alone. Second, clause-initial finite-verb mutation (`synt:v1-*`) is **not XPTH-derivable even in principle**: nothing precedes a clause-initial verb, and the negative variant surfaces as *mixed* mutation (AM on *c/p/t*) where the XPTH only ever produces SM. It is particle-drop residue — affirmative *fe°/mi°*, interrogative *a°*, negative *ni* (mixed) were deleted, and their mutations remained (King §481 note; Green 2003 on *ni*). The grade tracks the vanished particle exactly. In formal literary Welsh the affirmative default is particle-less and radical, so this entire rule is register-specific (King §11d): stated as designated clause positions, the difference between registers is the presence or absence of those positions, not a different theory. Imperatives resist both v1 mutation and *neu°* (King §512), and so constitute a category of their own.

Gwelodd Mair °dŷ
'Mair saw a house' — direct object mutation: the subject NP's right edge licenses, not the verb (Harlow 1989; Borsley 1997; Tallerman 2006)

```
S
├─0 gweld ⟨V⟩ → SM (synt:v1-aff)
├─1 NP
│  └─0 Mair ⟨N⟩ immutable → radical (no-license)
└─2 NP
   └─0 tŷ ⟨N m sg⟩ → SM (synt:xp-edge)
```

> The verb also carries colloquial v1 SM (°Welodd) — see the v1 examples below; the annotation on the verb reflects this.

Roedd dyn wedi prynu beic
'A man had bought a bike' — the object of a NON-finite verb stays radical: no phrase edge precedes it

```
S
├─0 bod ⟨V⟩ → radical (veto:no-reflex blocks synt:v1-aff)
├─1 NP
│  └─0 dyn ⟨N m sg⟩ → radical (no-license)
├─2 wedi ⟨Prt⟩ → radical (veto:no-reflex blocks synt:xp-edge)
└─3 VNP
   ├─0 prynu ⟨Vnoun⟩ → radical (no-license)
   └─1 NP
      └─0 beic ⟨N m sg⟩ → radical (no-license)
```

Roedd dyn wedi prynu yn y °dre °feic
'A man had bought, in town, a bike' — the decisive datum for the configurational account: an intervening PP restores mutation on the nonfinite object (Green 2003)

```
S
```

```
├─0 bod ⟨V⟩ → radical (veto:no-reflex blocks synt:v1-aff)
├─1 NP
│  └─0 dyn ⟨N m sg⟩ → radical (no-license)
├─2 wedi ⟨Prt⟩ → radical (veto:no-reflex blocks synt:xp-edge)
└─3 VNP
   ├─0 prynu ⟨Vnoun⟩ → radical (no-license)
   ├─1 PP
   │  ├─0 yn ⟨Other⟩ lemma=yn.loc → radical (no-license)
   │  └─1 NP
   │     ├─0 y ⟨Other⟩ → radical (no-license)
   │     └─1 tre ⟨N f sg⟩ → SM (gend:art-fem-sg)
   └─2 NP
      └─0 beic ⟨N m sg⟩ → SM (synt:xp-edge)
```

Beic °brynodd y °ddynes
'It was a bike the woman bought' — the fronted object has nothing before it and stays radical; a Case-based account must stipulate this (Tallerman 2006 contra Roberts 1997)

```
S
├─0 NP
│  └─0 beic ⟨N m sg⟩ → radical (no-license)
├─1 prynu ⟨V⟩ → SM (synt:xp-edge)
└─2 NP
   ├─0 y ⟨Other⟩ → radical (no-license)
   └─1 dynes ⟨N f sg⟩ → SM (gend:art-fem-sg)
```

Pwy °welodd °ddraig?
'Who saw a dragon?' — the extraction gap occupies the subject position and counts as a phrase edge; the verb's own SM (°welodd < gwelodd) comes from the fronted wh-phrase's edge

```
S
├─0 NP
│  └─0 pwy ⟨Other⟩ → radical (no-license)
├─1 gweld ⟨V⟩ → SM (synt:xp-edge)
├─2 gap:NP
└─3 NP
   └─0 draig ⟨N f sg⟩ → SM (synt:xp-edge)
```

Rhaid i Emrys °fynd
'Emrys must go' — the PP's right edge c-commands and licenses the verbal noun; King files this under 'sentence construction' (§5e: unblockable), for the XPTH it is the same rule as DOM

```
S
├─0 rhaid ⟨N⟩ → radical (no-license)
├─1 PP
│  ├─0 i ⟨Other⟩ → radical (no-license)
│  └─1 NP
│     └─0 Emrys ⟨N⟩ immutable → radical (veto:no-reflex blocks lex:i)
└─2 VNP
   └─0 mynd ⟨Vnoun⟩ → SM (synt:xp-edge)
```

°Golles i'r tocyn
'I lost the ticket' — DOM lands on the first WORD of the object; here that is the article (no SM reflex), so the noun stays radical

```
S
├─0 colli ⟨V⟩ → SM (synt:v1-aff)
├─1 NP
│  └─0 i.pron ⟨Other⟩ → radical (no-license)
└─2 NP
   ├─0 y ⟨Other⟩ → radical (veto:no-reflex blocks synt:xp-edge)
   └─1 tocyn ⟨N m sg⟩ → radical (no-license)
```

°Ddylset ti °ddim
'You shouldn't' — two mutations, two subsystems: mixed particle-residue on the negative verb, subject-edge SM on dim (King §§10, 11a)

```
S (neg)
├─0 dylu ⟨V⟩ → SM (synt:v1-neg-mixed)
├─1 NP
│  └─0 ti ⟨Other⟩ immutable → radical (no-license)
└─2 dim ⟨Prt⟩ → SM (synt:xp-edge)
```

Pharith hi °ddim
'It won't last' — the negative v1 grade is MIXED: AM on p-, so no soft mutation; the grade tracks the dropped particle ni (King §10)

```
S (neg)
├─0 para ⟨V⟩ → radical (no-license)
├─1 NP
│  └─0 hi ⟨Other⟩ → radical (no-license)
└─2 dim ⟨Prt⟩ → SM (synt:xp-edge)
```

os daw e
'if he comes' — the subordinator occupies clause-initial position, so the verb is not v1 and stays radical (King §502)

```
S
├─0 os ⟨Other⟩ → radical (no-license)
├─1 dod ⟨V⟩ → radical (no-license)
└─2 NP
   └─0 e ⟨Other⟩ → radical (no-license)
```

°Ddwy °flynedd yn ôl aeth hi adre
'Two years ago she went home' — adverbial NPs mutate at their first word (King §11b); note the structurally identical fronted object (dom-fronted) does NOT — the difference is adjunct-vs-argument status, not configuration

```
S
├─0 NP (adverbial)
│  ├─0 dwy ⟨Num⟩ → SM (synt:adv-np)
│  ├─1 blynedd ⟨N f sg⟩ → SM (lex:dwy)
│  └─2 yn ôl ⟨Adv⟩ → radical (no-license)
```

```
├─1 mynd ⟨V⟩ → radical (veto:no-reflex blocks synt:xp-edge)
└─2 NP
   └─0 hi ⟨Other⟩ → radical (no-license)
```

Dewch, °blant!
'Come, children!' — vocative mutation (King §11c); the imperative verb itself stays radical: imperatives resist v1 mutation

```
S
├─0 dod ⟨Vimp⟩ → radical (no-license)
└─1 NP (vocative)
   └─0 plant ⟨N m pl⟩ → SM (synt:vocative)
```

6. Vetoes

Two ways a word can be exempt no matter what licenses it: a per-lexeme immutability flag (King §12: fixed mutations like *beth*; personal names; g-initial loanwords; miscellaneous grammatical words), and the absence of any SM reflex for the word's initial (vowels, *n-*, *s-*, *ch-*, *ff-* …, King §5a). The two are distinguished throughout because they are different theoretical claims: one is lexical listing, the other phonological vacuity.

dy gêm
'your game' — g-initial loanwords are immutable even under a live trigger (King §12e)

```
NP
├─0 dy ⟨Other⟩ immutable → radical (no-license)
└─1 gêm ⟨N f sg⟩ immutable → radical (veto:immutable blocks lex:dy)
```

i Dafydd
'to Dafydd' — personal names do not mutate in the modern language (King §12d, §36)

```
PP
├─0 i ⟨Other⟩ → radical (no-license)
└─1 NP
   └─0 Dafydd ⟨N⟩ immutable → radical (veto:immutable blocks lex:i)
```

i ysgol
'to school' — vowel-initial words have no SM reflex; the question does not arise (King §5a)

```
PP
├─0 i ⟨Other⟩ → radical (no-license)
└─1 NP
   └─0 ysgol ⟨N f sg⟩ → radical (veto:no-reflex blocks lex:i)
```

7. Contested territory

A predictive account must say where prescription itself is unstable. The contexts below are catalogued as *contested*: the rules above state the prescriptive norm, and predictions in these environments should be read with that caveat. Corpus context: overall prescriptive application in spontaneous speech runs near 74%, no speaker is categorical, gender agreement on adjectives applies barely above chance (54%), and the particle-less finite-verb SM is the most categorical trigger measured (95%) — consistent with relexicalization of the mutated verb forms (Jones, n.d.; Knight et al. 2020).

- **euphony:s-d** — d- often fails to mutate after preceding -s: nos da, wythnos diwetha (King §12e note). King treats this as a euphonic override of the gender-agreement rule.
- **am-erosion** — AM after â/gyda/tri frequently realized as SM or radical colloquially (King p.12; corpus: gyda 2.94% AM). The prescriptive rule (AM ⇒ no SM) is stated here; SM in these contexts is attested vernacular.
- **immutable-variation** — lle, byth: immutability 'usage varies' (King §12b).
- **v1-neg-sm** — Negative inflected verbs prescriptively take mixed mutation, but plain SM is common speech (King §10: 'more a feature of the literary language'). The mixed grade is stated here.
- **placenames** — Non-Welsh placenames prescriptively immutable (i Birmingham), but i Firmingham 'not uncommon' in speech (King §12c). After yn 'in', foreign names 'even more chaotic': yng Nghamden / yn Gamden / yn Camden all heard (King §472).
- **nm-blynedd** — NM of blynedd/blwydd after chwe and wyth 'sometimes followed by the radical': chwe blynedd, wyth blynedd (King §176 note a).
- **yn-loc-sm** — NM after yn 'in' is 'precarious at best' in speech; if any mutation heard it is usually SM (yn °Fangor, yn °Geredigion), though SM of G- names is resisted (yn Gogledd Cymru) and bare radical is common (yn Bangor). Formal written language allows none of these (King §472). Fixed exception either way: yn °Gymraeg takes SM even in the literary language (blocked NM from a lost article).
- **tan-conj** — tan as a conjunction (tan iddo ffonio) 'certainly used' in some areas but 'some speakers regard this as substandard'; standard is nes (King §467).
- **cyn-finite** — cyn + inflected verb (cyn daeth e adre) 'regarded as substandard' for cyn iddo °ddod adre (King §509).
- **tra-grade** — Degree adverb tra: King marks tra° (SM, §95); the literary tradition prescribes AM (tra chyflym). King's SM is stated here.
- **am-dialect-mn** — In some dialects AM extends to radical m-/n- (m > mh, n > nh), 'regarded as non-standard' (King App A p.488). Irrelevant to soft mutation itself, but a hazard for analysis: mh-/nh- surface forms are not always NM.

8. The full trigger lexicon

The complete inventory of contact-trigger conditions, with King references. A single word may govern several distinct conditions (*y*, *chwe*); grade `none` entries record deliberate non-triggers.

Lemma	Grade	Target conditions	Source
`am`	SM	—	§448
`ar`	SM	—	§449
`at`	SM	—	§450
`cyn.prep`	none	—	§451 (no superscript; contrast cyn- prefix and cyn.equ)
`dan`	SM	—	§452
`o dan`	SM	—	§452
`oddidan`	SM	—	§452
`dros`	SM	—	§453

Lemma	Grade	Target conditions	Source
`tros`	SM	—	§453 (literary variant)
`drost`	SM	—	§453 (spoken variant)
`efo`	none	—	§454 (N 'with')
`gan`	SM	—	§455
`gyn`	SM	—	§455 (spoken variant of gan)
`ger`	none	—	§456
`gerbron`	none	—	§456
`gyda`	AM	—	§457; erosion CONTESTED (am-erosion)
`heb`	SM	—	§458
`hyd`	SM	—	§459
`i`	SM	—	§460
`o`	SM	—	§9
`tan`	SM	—	§467
`trwy`	SM	—	§9
`drwy`	SM	—	§9
`wrth`	SM	—	§9
`â`	AM	—	§447 ('optionally causes AM') CONTESTED (am-erosion)
`tua`	AM	—	App A
`yn.loc`	NM	—	App A, §472
`a.rel`	SM	—	§481
`dacw`	SM	—	App A
`dyma`	SM	—	App A
`dyna`	SM	—	App A
`fe.prt`	SM	—	App A §213
`mi.prt`	SM	—	App A §213
`yma`	SM	—	App A
`yna`	SM	—	App A
`neu`	SM	cat ∈ {N, Adj, Vnoun, Num, Adv, Prt, Other}	§512; SM cancelled before an imperative (Arhoswch neu dewch) — hence V excluded
`pan`	SM	—	App A
`yn.pred`	SM-ltd	cat ∈ {N, Adj, Num}	App A ('not ll-, rh-'); numerals in age/time: yn °dair oed §176
`go`	SM	—	§95
`pur`	SM	—	§95
`rhy`	SM	—	§95
`tra`	SM	—	§95 (King marks tra°; literary AM tradition) CONTESTED

Lemma	Grade	Target conditions	Source
`reit`	SM	—	§95
`mor`	SM-ltd	—	§105(e), App A
`cyn.equ`	SM	—	§105 (cyn °ddued)
`na.than`	AM	—	§103 (na^{h}, nag before vowels)
`fatha`	AM	—	§105 (colloquial ‘like’)
`fawr`	SM	—	§105(d) (fawr °gallach)
`lawn`	SM	—	§105(a) (°lawn mor °grac)
`y`	SM-ltd	cat ∈ {N}; gender=f; number=sg	§28; note (b) ll-/rh-resist
`y`	SM	cat ∈ {Num}; lexemes: dau/dwy	§29 (y °ddau, y °ddwy)
`un`	SM-ltd	cat ∈ {N}; gender=f; number=sg	§162(a)
`fy`	NM	—	§110
`dy`	SM	—	§111
`ei.3sgm`	SM	—	§112
`ei.3sgf`	AM	—	§112; colloquial omission CONTESTED (am-erosion)
`ein`	none	—	§113 (h- before vowels only)
`eich`	none	—	§113
`eu`	none	—	§113 (h- before vowels only)
`'w.3sgm`	SM	—	§112 (i’w °frawd)
`'w.3sgf`	AM	—	§112
`'w.3pl`	none	—	§112
`hen`	SM	—	§96(a)
`prif`	SM	—	§96(a)
`ambell`	SM	—	§96(a), §116
`holl`	SM	—	§96(a)
`pob`	none	—	§97 (the ONLY prenominal adjective that does not trigger SM)
`pa`	SM	—	§96(b)
`pwy.S`	SM	—	§96(b) (S variant of pa)
`cyn-`	SM	—	§96(c) (ex-; prefix written with hyphen but treated as trigger word)
`dirprwy`	SM	—	§96(c)
`uwch`	SM	—	§96(c)
`is-`	SM	—	§96(c)

Lemma	Grade	Target conditions	Source
`cryn`	SM	—	§96(d)
`unig`	SM	—	§99 (prenominal sense ‘only’; postnominal ‘lonely’ does not trigger)
`rhyw`	SM	—	§115
`unrhyw`	SM	—	§115
`amryw`	SM	—	§116
`cyfryw`	SM	—	§116 (y cyfryw °beth)
`fath`	SM	—	§116 (y fath °beth)
`ffasiwn`	SM	—	§116
`rhai`	none	—	§115 (contrast rhyw°)
`sawl`	none	—	§187
`peth`	none	—	§193 (directly precedes sg noun, no o°, no mutation)
`dau`	SM	—	§162(b)
`dwy`	SM	—	§162(b)
`tri`	AM	—	§162(c) ‘erratically in the spoken language’ CONTESTED (am-erosion)
`tair`	none	—	§162(c) (always radical)
`pedwar`	none	—	§162
`pedair`	none	—	§162
`chwe`	AM	—	§162(e) ‘erratically in speech’ CONTESTED (am-erosion)
`chwe`	NM	lexemes: blwydd/blynedd/diwrnod	§176; radical sometimes (chwe blynedd) CONTESTED (nm-blynedd)
`pum`	NM	lexemes: blwydd/blynedd/diwrnod	§176
`saith`	NM	lexemes: blwydd/blynedd/diwrnod	§176
`wyth`	NM	lexemes: blwydd/blynedd/diwrnod	§176; radical sometimes (wyth blynedd) CONTESTED (nm-blynedd)
`naw`	NM	lexemes: blwydd/blynedd/diwrnod	§176

Lemma	Grade	Target conditions	Source
deng	NM	lexemes: blwydd/blynedd/di-wrnod	§176
deuddeng	NM	lexemes: blwydd/blynedd/di-wrnod	§176 (12)
pymtheng	NM	lexemes: blwydd/blynedd/di-wrnod	§176 (15)
deunaw	NM	lexemes: blwydd/blynedd/di-wrnod	§176 (18)
ugain	NM	lexemes: blwydd/blynedd/di-wrnod	§176 (20)
can	NM	lexemes: blwydd/blynedd/di-wrnod	§176 (100: can °mlwydd oed)
ail	SM	—	§170(b) (both genders)
trydedd	SM	gender=f	§170(b)
pedwaredd	SM	gender=f	§170(b)
trydydd	none	—	§170(b) (masc: y Trydydd Byd)
pedwerydd	none	—	§170(b)
pumed	SM	gender=f	§170(b) (y °bumed °orsaf; masc radical)
chweched	SM	gender=f	§170(b)
seithfed	SM	gender=f	§170(b)
wythfed	SM	gender=f	§170(b)
nawfed	SM	gender=f	§170(b)
degfed	SM	gender=f	§170(b)
ugeinfed	SM	gender=f	§170(d) (yr ugeinfed °ganrif)
nos	SM	lexemes: Llun/Mawrth/Mercher/Iau/Gwener/Sad-wrn/Sul/Calan	§180 (Nos °Fawrth, Nos °Galan)
mewn	none	—	§461 (non-specific 'in a'; contrast yn.loc)
mo	SM	—	§462, §295 (contraction of dim o°, carries o's SM)
oddiar	SM	—	§463
oddiwrth	SM	—	§464
rhag	none	—	§465 (rhag dod, rhag ofn — radical)

Lemma	Grade	Target conditions	Source
`rhwng`	none	—	§466 (‘one of the few simple spatial prepositions which does not cause SM’)
`trw`	SM	—	§468 (variant of trwy)
`wth`	SM	—	§470 (spoken variant of wrth)
`er`	none	—	§506 (er ei holl gyfoeth — no mutation from er itself)
`ers`	none	—	§503
`erbyn`	none	—	§503
`nes`	none	—	§503 (i-construction; no direct mutation)
`ar bwys`	none	—	§475 (compound preps cause no mutation on following noun)
`ar draws`	none	—	§475
`ar gyfer`	none	—	§475
`ar ôl`	none	—	§475
`er mwyn`	none	—	§475
`o flaen`	none	—	§475
`o gwmpas`	none	—	§475
`wrth ymyl`	none	—	§475
`ymhlith`	none	—	§475-476
`ymysg`	none	—	§475-476
`yn lle`	none	—	§475
`hanner can`	NM	lexemes: blwydd/blynedd/di-wrnod	§176 (50)
`a.conj`	AM	—	§510; ‘often disregarded in the spoken language, particularly for th-’ CONTESTED (am-erosion)
`na.nor`	AM	—	§513 (nor; ‘often disregarded in speech’) CONTESTED (am-erosion)
`na.rel`	mixed	—	§481 (NEG relative; King’s exx show SM on b/d/g-initials, consistent with mixed; AM on c/p/t per prescriptive tradition + §10)

Lemma	Grade	Target conditions	Source
`sy`	SM	lexemes: dim	§479 (sy °ddim)
`os`	none	—	§502 (os daw e)
`pe`	none	—	§502
`hyd.conj`	none	—	§502 (hyd gwelwch chi — verb radical; contrast hyd° preposition)
`ta`	none	—	§512 (te 'ta coffi)
`ond`	none	—	§511
`ni`	mixed	—	§10
`na.neg`	mixed	—	§10
`oni`	mixed	—	§10

9. Known limits

- **Adverbial and vocative status must be given, not derived.** Constituent geometry provably underdetermines them: sentence-initial adverbial NPs mutate while structurally identical fronted objects do not. The distinction is adjunct versus argument — a valence/lexical fact every formal account takes as input (Dowle 2024 reads it from f-structure); none derives it configurationally.
- **Common-noun apposition** shares the genitive configuration [NP N NP] and is therefore predicted, wrongly, to pattern with possessors; appositive NPs are overwhelmingly personal names (immutable regardless), so the exposure is minimal.
- **Coordination** is a known soft spot for the XPTH generally; Welsh coordinators happen to begin with mutation-immune segments (*a*, *neu*, *ond*, *na*), which masks most of the exposure.
- **Usage variability** (§7) is outside predictive scope by design; a usage model would attach per-trigger application rates to the same rule inventory.
- The mixed-mutation grade of the negative relative particle *na* follows prescriptive tradition; King's own examples (§481) are grade-ambiguous (all on *b*/*d*/*g*-initials, where mixed and SM coincide).

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